

# Phonetics Assignment

## Acoustic Analysis of Vowel Quality Using Formant Extraction

**Submission format: Report (PDF) + Scripts + Audio files**

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### 1. Learning Objectives

By completing this assignment, you will:

- Record high-quality speech data suitable for acoustic analysis
  - Extract vowel formant values (F1, F2, and F3 if needed) using automated scripts
    - Praat scripts are available on Arche but you're welcome to write something yourself in your prog language of choice
  - Visualize vowel spaces for individual speakers
    - R scripts are available on Arche but you're welcome to write something yourself in your prog language of choice
  - Compute and interpret basic descriptive statistics of vowel quality
    - Mean, median, standard deviation, vowel space area
  - Critically evaluate the results and methodological limitations in acoustic vowel analysis
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### 2. Task Overview

You will:

1. Record approximately **10 sentences** containing multiple tokens of target vowels.
2. Segment and label sentences, words, and phones intervals.
  1. You can do it either manually or automatically
  2. You can use material you recorded for other speech classes
3. Extract **F1 and F2 values** (F3 if needed for rounded vowels) from the audio signal using scripts (e.g., Praat, Python, or R).
4. Plot vowel spaces for each speaker using scripts (e.g., Praat, Python, or R).
5. Compute basic statistics describing each vowel category as well as the overall vowel spaces
6. Write a short report interpreting your findings.

You may work individually or in groups (maximum 4 students per group).

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### 3. Speech Recording

#### 3.1 Materials

Prepare approximately **10 phonetically balanced sentences** containing multiple vowels.

Requirements:

- Include at least **3–8 vowel categories** of the language.
- Each vowel category should appear at least **3 times**.
- Prefer stressed syllables in controlled phonological environments (e.g., /hVd/, /bVt/, etc., if applicable to the language).

You may:

- Construct your own sentences
- Or adapt from established materials (cite your source / the class you did the work for)

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#### 3.2 Recording Conditions

- Record in a **quiet room**
- Use a **head-mounted or high-quality external microphone**
- Record in **mono**
- Sampling rate : **22 050 Hz**
- Format: **WAV (not MP3 – not AIFF / any proprietary format)**
- Record each speaker in a separate file
- Avoid clipping and background noise

Each group must include at least **4 speakers** (male/female comparison encouraged but not required).

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### 4. Segmentation and Annotation

Using Praat or similar software:

1. Create a **TextGrid** file.
2. Segment and label each sentence, word, phone.
3. Label sounds using SAMPA or X-SAMPA symbols.
4. Mark vowel boundaries carefully :
  - Onset: beginning of periodicity / clear formant structure
  - Offset: end of periodicity or before consonantal closure

Save:

- WAV file
  - TextGrid file
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## **5. Formant Extraction**

### **5.1 Measurement Specifications**

Extract:

- F1
- F2
- F3 (only if you have rounded and unrounded vowels)

Measure:

- At vowel midpoint (50% of duration)
- And/or at each 1/3 of the vowel duration

Report:

- Formant values in Hz
  - Duration (optional but recommended)
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### **5.2 Script-Based Extraction**

You must use an automated script in:

- Praat
- Python (e.g., Parselmouth, Librosa)
- R (e.g., wrassp, phonTools, PhonR)

Your script must:

- Read labeled intervals
- Extract midpoint and/or several points for F1 and F2 (F3 if necessary)
- Export results as CSV

Submit your script file with clear comments.

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## **6. Data Visualization: Vowel Spaces**

Using R, Python, or another statistical tool:

1. Plot F1 (y-axis) vs. F2 (x-axis)

2. Reverse axes according to phonetic convention:

- F1: high vowels at top
- F2: front vowels on the left

3. Plot:

- Individual vowel tokens
- Category means

4. Produce:

- One vowel space per speaker
- Optional: overlay comparison plot

Label all axes clearly (Hz).

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## 7. Statistical Analysis

For each speaker and each vowel category compute:

- Mean F1
- Mean F2
- (Mean F3)
- Standard deviation for F1 F2 (optionally F3)
- Number of tokens
- Vowel space area (convex hull)
- Euclidean distances between selected vowel pairs

Present results in clearly formatted tables.

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## 8. Report Structure (about 2000-3000 words, single spaced, 12 pt font, Times New Roman, standard margins)

Your report must include:

### 1. Introduction

- Brief theoretical background on vowel acoustics
- Relationship between formants and vowel quality (acoustics/articulatory relationship)

### 2. Method

- Speakers
- Materials

- Recording conditions
- Segmentation procedure
- Formant extraction settings

### 3. Results

- Vowel space plots
- Statistical tables
- Description of patterns

### 4. Discussion

Discuss:

- Differences between speakers
- Vowel dispersion
- Centralization or overlap
- Effects of context (if observed)
- Measurement limitations (e.g., formant tracking errors)

### 5. Conclusion

Brief summary of key findings.

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### 9. Submission Checklist

Submit the following:

- WAV files
- TextGrid files
- Extraction script
- CSV output
- Vowel space plots (PDF or PNG)
- Final report (PDF)

All files must be organized in clearly labeled folders.

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### 10. Evaluation Criteria

| Component                   | Weight |
|-----------------------------|--------|
| Data quality & segmentation | 20%    |

| Component                      | Weight |
|--------------------------------|--------|
| Script functionality           | 15%    |
| Accuracy of formant extraction | 15%    |
| Quality of visualizations      | 15%    |
| Statistical analysis           | 15%    |
| Interpretation & discussion    | 20%    |